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A lightweight georeferencing workflow for dynamic UAV footage using feature-matching and minimal drone metadata

Muhammad Waqas Ahmed^{a,*}, Muhammad Adnan^a, Muhammad Ahmed^b,
Davy Janssens^a, Geert Wets^a, Afzal Ahmed^c, Wim Ectors^a

^a UHasselt, Transportation Research Institute (IMOB), Martelarenlaan 42, 3500, Hasselt, Belgium

^b Department of Urban and Infrastructure Engineering, NED University of Engineering and Technology, Karachi, Pakistan

^c Institute of Transportation Studies, University of Leeds, Leeds, UK

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ABSTRACT

The emergence of unmanned aerial vehicles (UAVs), commonly known as drones, has transformed aerial imaging and photogrammetry, offering a cost-effective and flexible alternative to traditional methods. While commercially available drones are useful and affordable, the metadata provided in flight logs often falls short for robust photogrammetric applications. To address this limitation, we propose a novel method for the automated georeferencing of UAV footage that combines a feature-matching algorithm, Scale Invariant Feature Transform (SIFT), with telemetry data. Our system begins by initializing the homography by matching an input frame with a reference orthomosaic with a known spatial projection. Subsequently, the homography of the following frames is adjusted using the translation component estimated from the drone's telemetry. For the drone's rotation, the Oriented FAST and Rotated BRIEF (ORB) algorithm was utilized to detect changes between consecutive frames, allowing for reinitialization of the homography when needed. To quantify uncertainty and assess temporal dependence in frame-wise accuracy, a moving-block bootstrap (MBB) approach was employed for estimating confidence intervals. The proposed workflow is designed to be modular, meaning that the algorithms can be swapped out based on the data and conditions. Experimental results indicate that the method achieves sub-meter accuracy, with mean RMSE ranging from 54.9 to 95.9 cm.

1. Introduction

The emergence of unmanned aerial vehicles (UAVs) or drones marked a paradigm shift in aerial imaging and photogrammetry, offering a low-cost and scalable alternative to conventional airborne imaging methods. Drones boast exceptional mobility and the ability to access hard-to-reach areas, which gives them a significant advantage. This has led to their widespread use across various fields, including agrotechnology, earth observation, geological exploration, and climate-related applications (Tewes, 2018). In photogrammetry, UAVs offer high-resolution imaging at a lower operational cost. The evolution of drone technology has resulted in several types of drones, including fixed-wing UAVs, single-rotor drones, multirotor (Quadcopters, Hexacopters, and Octocopters), and fixed-wing VTOL UAV systems (Sönmez et al., 2022). The vertical take-off and landing (VTOL) capabilities of hybrid drones minimize the need for extensive infrastructure, with longer range yet as maneuverable as multirotor drones (Colomina and Molina, 2014).

* Corresponding author.

E-mail address: Waqas.ahmed@uhasselt.be (M.W. Ahmed).

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Precise georegistration or georeferencing of UAV-based aerial imagery is crucial for photogrammetric applications. This process has a wide range of applications, including 3D scene reconstruction (Pierrot-Deseilligny et al., 2011), land use planning (Liu et al., 2018), and precision agriculture (González-García et al., 2020). In today's age of artificial intelligence (AI), accuracy in georeferencing also contributes to effectively geolocating objects in real-world contexts, such as road traffic and tracking vehicle movements. This capability can enable automated workflows for road traffic monitoring (RTM) (Ahmed et al., 2024).

Photogrammetry applications go beyond traditional surveying applications. Image georegistration is a fundamental step for downstream applications, such as change detection, feature extraction, object detection, and localization (Doucette et al., 2013). These downstream applications benefit from correctly georeferenced images. However, the standard telemetry provided by commercially available drones only includes center coordinates and lacks the necessary parameters to map the captured imagery into the real geographical space. To enable automated georeferencing, vision-based methods, such as feature-matching algorithms for automated image georegistration and matching, are highly effective (Zhuo et al., 2017). The main drawback of vision-based methods is that applying them to every frame is practically infeasible due to high computational costs. Additionally, as the drone moves, the varying keypoints across different locations can lead to performance inconsistencies.

Frame-by-frame SIFT implementation is computationally heavy; therefore, downstream applications become practically unfeasible, as demonstrated in the authors' previous research (Ahmed et al., 2024). To avoid frame-by-frame feature matching, this paper proposes a hybrid method that combines the drone's positional data with SIFT feature matching. The system only requires latitude, longitude, and altitude information from the drone's telemetry, which simplifies the requirements for automated georeferencing applications, ensuring both speed and precision. In the proposed method, the homography of the first video frame is initialized. The translation terms are updated based on geodesic displacements (Δx , Δy) relative to a fixed reference to track positional changes over time using UAV telemetry. These changes are converted into pixel shifts to adjust the image's position while keeping its orientation and scale intact. The framework tackles the lack of orientation information in the drone's telemetry data by using the ORB (Oriented FAST and Rotated BRIEF) algorithm for rotational detection. If rotation is detected, the homography is reinitialized to ensure that the output is spatially accurate.

2. Related works

Conventional UAV systems typically feature lightweight GNSS and IMU systems, and the efficiency of different hardware components can vary (Jacobsen et al., 2010). Chiabrando, Lingua (Chiabrando et al., 2013) assessed the performance of onboard GNSS/INS systems for direct georeferencing. In this study, the authors utilized three different UAV platforms and the block bundle adjustment (BBA) method for orthophoto generation. They compared the accuracy of each platform's output with ground control points from a total station. The experimental findings indicate that while lightweight GNSS and IMU systems may be effective for medium-scale mapping, they pose significant limitations for large-scale photogrammetry, which demands a centimeter-level accuracy. Although advanced hardware can considerably improve the accuracy, it also increases costs and may introduce regulatory hurdles due to an increase in maximum take-off mass (MTOM). To counter this limitation, researchers have explored the viability of pre-georeferenced aerial or satellite imagery as a reference for feature-matching-based methods for direct registration. In their study, Zhuo, Koch (Zhuo et al., 2017) presented a method for automated georegistration of UAV images, utilizing a dense feature detection scheme. The authors emphasize the accuracy of their approach, highlighting its utility over GNSS-based methods. However, the authors did not discuss the computational expense and scalability of their proposed solution. Another similar study by Bas and Ok (Bas et al., 2021) presents AKAZE (Accelerated-KAZE) local descriptors for matching oblique aircraft and UAV-based images. The proposed framework performs well; however, it is essential to acknowledge the limitations of AKAZE, specifically its sensitivity to scale (Bas et al., 2021; Tareen and Saleem, 2018). Similarly, feature-matching algorithms can be used to develop a fully automated solution for georeferencing. Li, Zhang (Li et al., 2024) in their study presented a solution by leveraging positional and orientation system (POS) data for ground range and resolution estimation, followed by a SuperGlue network for feature matching. Geometric rectifications were performed using a third-order polynomial transformation, and the results demonstrate accuracy ranging from 2 to 5 m. Despite the automated nature of the workflow, its primary limitation is the per-frame application of feature-matching algorithm, which our proposed work aims to alleviate. Recent trends have been more focused on learning-based algorithms. One such study was conducted by Zhu, Ye (Zhu et al., 2024), where the authors introduced a Viewpoint-invariant Deformable Feature Transformation (VDFT) for aerial-to-ground feature matching, addressing challenges posed by viewpoint and illumination differences. VDFT includes a deformable feature network for geometric variations, an improved detection strategy for localization accuracy, and a seed attention module for enhanced matching performance. Experimental results on various datasets demonstrate that VDFT outperforms state-of-the-art methods in robustness and accuracy. While learning-based algorithms are generally more robust and efficient than traditional algorithms, their effectiveness is ultimately constrained by the quality of the data on which they have been trained.

Apart from georeferencing applications, feature-matching algorithms are also used for positional corrections of UAVs in GPS-denied environments. The key principle behind feature point detection is finding stable key points immune to scaling, rotation, and varying pixel intensities. The primary objective of feature point matching is to construct a descriptor to estimate Euclidean or Hamming distance. Scale-invariant feature transform, or SIFT, is the pioneering algorithm in this category, developed by Lowe (2004). Continued research in this domain resulted in other noteworthy algorithms, such as speeded-up robust features (SURF) (Bay et al., 2008), binary robust independent elementary features (BRIEF) (Calonder et al., 2010), and oriented FAST and rotated BRIEF (ORB). Qu, Sohelian (Qu et al., 2016) evaluated the robustness of SIFT with its faster counterpart, i.e., SURF, and compared both on the basis of stability, variance of points in image space, localization accuracy, and computation cost. The experimental results confirmed that although SURF was faster than SIFT, it lacked reliability and precision. The results also confirm that image resolution is key to

localization accuracy. Another experiment conducted by Tanchenko, Fedulin (Tanchenko et al., 2020), demonstrated the power of SIFT to perform positional corrections for road and river network extraction. The proposed workflow acquires a reference image and performs segmentation to delineate the road and river networks. A feature vector is then constructed using the SIFT method, followed by map matching and calculating a homography, which is a 3×3 matrix used for planar transformations. This process enables an image to be converted into the projection of its corresponding reference frame, allowing for the application of positional corrections.

The computational expense of SIFT has been a major hindrance to its implementation in automated workflows. Researchers have attempted to address this problem by incorporating different techniques. Ke and Sukthankar (Ke and Sukthankar, 2004) presented an optimized version of SIFT by implementing Principal Component Analysis (PCA). They used the standard SIFT descriptor and applied principal PCA to the 128-dimensional vector. The authors state that this high number of dimensions is unnecessary to describe a key point; therefore, reducing the dimensionality while retaining essential information can decrease the processing time. Another similar study employed PCA-SIFT method and further enhanced its speed by limiting the keypoint search during a UAV flight (Van Quan et al., 2023). In the experiment, the UAV telemetry was used to limit the keypoint generation extent on the reference map. The approximation was done by using the UAV's position and comparing image dimensions and ground sampling distances to estimate the distance to the top-left geographical corner of the frame. Assuming that the distance remains the same across all four corners, a bounding box is created to limit the keypoint search on the reference map. This also reduces the chances of errors from bad matches since the algorithm is guided to match only a specific portion of the reference map. The benchmarking indicates that their version of PCA-SIFT, with a restricted search area, was 227 times faster than the standard SIFT.

Although the implementation of PCA to SIFT descriptors and limiting the search area for keypoints search does improve the processing speed, changes in the position of the UAV within the input footage mean that a different set of keypoints will be selected for matching, resulting in divergent outcomes for each frame. Consequently, restricting the number of SIFT-based homography corrections offers an even more efficient solution. Our proposed research demonstrates that by updating the translation terms for the frames in between using telemetry and applying SIFT when necessary (as detailed further in section 3.2).

3. Data and methods

The proposed methodology employs a minimal set of telemetry parameters obtained from a commercially available UAV, specifically the DJI Mini 3 Pro. The telemetry data was sourced from the drone's embedded (SRT) subtitle file, which contains latitude, longitude, altitude, focal length, and zoom ratio, recorded at a rate of 29.97 frames per second (FPS). We extracted latitude, longitude, and altitude, from the SRT files using a custom Python-based parser developed with standard module "re" for regular expression-based parsing of structured fields.

While positional data in the SRT file is available at a sufficient frequency, it does not account for UAV rotational dynamics. To address this limitation, the proposed workflow introduces a hybrid georeferencing approach that integrates UAV telemetry with image-based feature matching. This fusion enables robust frame-to-map alignment by combining external orientation data with visual

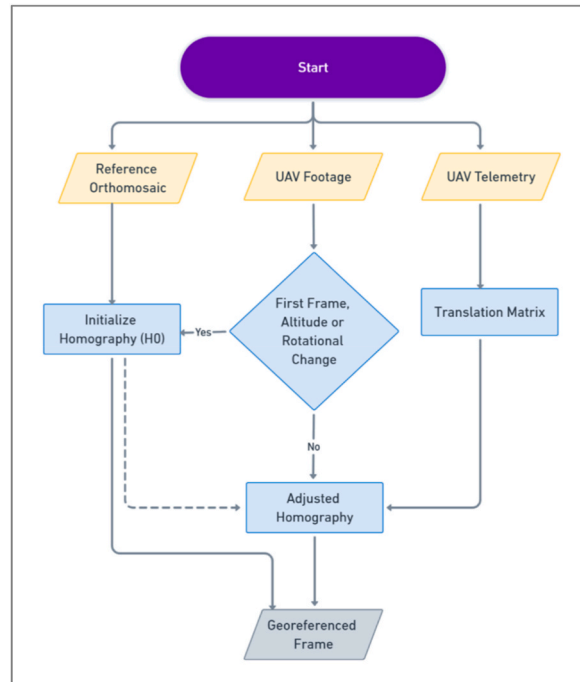


Fig. 1. Flowchart of the proposed methodological framework.

correspondences.

As illustrated in Fig. 1, the workflow integrates two independent data sources—video frames and telemetry data and produces a unified georeferenced output. Initially, the UAV footage is aligned with a georeferenced map or orthomosaic (projected in UTM Zone 31N) using a feature-matching algorithm (SIFT), which computes the initial homography matrix. This matrix serves as the reference or base homography. Subsequently, the translation component of the homography is updated on a per-frame basis to reflect UAV positional shifts from the origin. The overall workflow is implemented using two algorithms:

- **Algorithm 1** performs homography initialization using feature matching on the first frame and extends it across subsequent frames by incorporating telemetry-derived positional updates.
- **Algorithm 2** accounts for UAV rotational variations and adjusts the homography accordingly.

During frame processing, the workflow reinitializes the homography whenever the altitude change or rotational variation exceeds the predefined thresholds (see algorithm 1 in section 3.3). The final output consists of georeferenced frames generated in a fully automated manner using a minimal set of parameters from the telemetry, combined with image-based ground referencing.

3.1. Data acquisition and preparation

To evaluate the proposed method, two video sequences were collected at 4k resolution in Hasselt, Belgium, using a DJI Mini Pro 3 (discussed in Table 1). Across the two videos, Footage 01 consists of 2200 frames covering a total linear distance of 304.67 m, with the altitude varying from 55.7 m to 60.4 m and 67 instances of altitude change. Footage 02 contains 2000 frames over 168.4 m, with altitude ranging narrowly from 76.3 m to 76.5 m and 34 altitude change instances, which indicates that Footage 01 experienced more frequent and larger altitude variations, while Footage 02 remained relatively stable.

For the reference map, an orthomosaic (6065 × 4467) was created using WebODM (version 2.5.0) from a prior flight conducted at an altitude of 90 m. This reference imagery is projected in EPSG:32631 (WGS 84/UTM Zone 31N) with units in meters and a spatial resolution of 0.042 m/pixel. Algorithm 1 is designed to account for changes in altitude, so gathering footage with sufficient diversity was essential. The flight was conducted in compliance with aviation regulations, maintaining a clear visual line of sight (VLOS), and in full adherence to the General Data Protection Regulation (GDPR) (Rauhala et al., 2023; Bassi, 2020). In previous research (Ahmed et al., 2025), the efficacy of SIFT was tested under varying lighting conditions and angular changes. Given that 2D feature matching is highly sensitive to angular variations, footage was collected from a top-down or nadir angle during daylight hours. This approach was adopted to ensure that the feature matching algorithm operates effectively with two-dimensional images.

3.2. Feature matching using SIFT

The method utilizes feature-matching for homography initialization, and the SIFT algorithm was chosen due to its scale invariance. Unlike deep learning-based models, SIFT does not require pre-trained weights, which keeps the workflow straightforward and reliable. SIFT was initially introduced for general-purpose feature matching by Lowe (2004) in 2004. It was designed to detect distinctive invariant features from images, allowing for reliable matching between different views of an object or scene (see Fig. 2). The key advantage of SIFT lies in its ability to maintain invariance to factors, including image scale, rotation, illumination changes, 3D viewpoint alterations, affine distortion, noise addition, and blurring.

The SIFT process can be divided into four stages. The first stage is scale-space extrema detection, where a Gaussian scale-space pyramid is created. This involves generating successive blurred images through convolution with Gaussian functions, thus forming multiple octaves. The parameter sigma establishes the initial Gaussian blur for the first level of each octave. In this stage, the Difference of Gaussian (DoG) method is applied, calculating the difference between images at different scales within an octave. The parameter n-scale indicates the number of scales sampled in each octave. While increasing the number of scales can result in more detected key points, it may also decrease their stability. A balance must be struck between the total number of key points and the stable key points found. The second step is feature localization, where a detailed fitting is conducted for each extremum to ascertain its location, scale, and the ratio of principal curvatures. Keypoints found in areas of low contrast or along edges are filtered out to ensure the robustness of the feature set. Next is the orientation assignment phase, where an orientation is assigned to each keypoint based on the local image gradients. This step is crucial for achieving rotational invariance in the final feature descriptors. Finally, during the feature descriptor creation stage, SIFT computes a descriptor for each keypoint that remains invariant to residual variations. A keypoint descriptor is created by examining a 16x16 neighborhood around the keypoint. This area is divided into 16 sub-blocks, each measuring 4x4. For every sub-block, an 8-bin orientation histogram is generated, resulting in a total of 128 bin values. These values are then represented as a vector to form the keypoint descriptor (Castillo-Carrión and Guerrero-Ginel, 2017). SIFT has greater tolerance to seasonality due to its ability to match features despite variations in pixel intensities caused by variations in illumination or other seasonal causes, but the

Table 1
Specifications of the collected drone data.

Footage	Frames	Total Distance Traveled (m)	Start – Finish Altitude (m)	Altitude Change Instances
Footage 01	2200	304.67	55.70–60.40	67
Footage 02	2000	168.4	76.30–76.50	34



Fig. 2. Illustration of homography initialization through the feature point matching algorithm known as SIFT. The final transformed image (d) is overlaid on Google Earth imagery.

computational time for SIFT is extremely high, making it unsuitable for near real-time applications (Van Quan et al., 2023). Therefore, we propose a hybrid approach with drone telemetry to automatically optimize the georegistration of UAV frames.

In this experiment, we use the initial homography H_0 calculated by SIFT implementation as a control point and adjust the subsequent translation changes for each frame as $H_{adjusted}$. This approach reduces the need to perform SIFT matching for every frame. After detecting and computing keypoints using SIFT, PCA was applied to reduce the dimensions of the descriptors from 128 to 32, as demonstrated by Ke and Sukthankar (Ke and Sukthankar, 2004). It is noteworthy that the optimal number of dimensions may vary depending on the dataset; an overly aggressive reduction may lead to loss of discriminative information and introduce errors in H_0 . To ensure the reliability of these matches, Lowe's ratio test was applied to discard ambiguous keypoints. Subsequently, the Random Sample Consensus (RANSAC) method with reprojection threshold of 5.0 pixels and confidence level of 0.995 was applied to the key points for the removal geometric outliers and robust estimation of the homography (Kaplan et al., 2016). Our configuration used the default SIFT sigma value of 1.6 and three scales per octave to ensure stable keypoint detection.

3.3. Homography estimation and adjustment

The first step of the proposed method is to initialize a homography for the first frame through feature-matching. Images taken from various perspectives, altitudes, and sensors often show geometric and photometric variations due to lighting conditions and differences in camera technology (Luo et al., 2023). In order to align two images into a spatially identical projection system, it is important to apply a homography transformation with an orthorectified reference imagery with known geographical projection.

$$H_0 = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \quad (1)$$

Homography is represented by a 3x3 matrix (see equation (1)), which defines the relationship between points in two planes where $[h_{11}, h_{12}, h_{21}, h_{22}]$ describe the affine transformation, $[h_{31}, h_{32}]$ characterize the perspective transformation, and $[h_{13}, h_{23}]$ represent translation (Guan et al., 2018). Traditional methods for homography estimation involve three main steps: feature detection, feature matching, and solving the homography matrix. Feature extraction algorithms, such as SIFT, SURF, BRIEF, and ORB, are used to identify feature points and generate descriptors represented as vectors. Similarity measures like Hamming or Euclidean distance help match these feature points, while robust estimation techniques like RANSAC enhance the reliability of the homography matrix estimation (Luo et al., 2023).

To enable positional shifts, the translation component of the homography matrix is adjusted according to the geodesic distances $\Delta x = x - x_a$ and $\Delta y = y - y_a$. Here, we take our current "world" position (x, y) and a fixed reference or origin (x_a, y_a) in meters and compute East-North displacement. We then convert meter-scale shift into pixel offsets by dividing by the pixel size from the

geotransform. Each pixel maps to fixed ground distances in easting/northing, where p_x is the pixel width and p_y is the pixel height in meters. These cell sizes (p_x and $p_y = 0.042$ m/pixel), extracted from the reference image, are assumed constant across the frame (negligible lens-induced scale variation). The resultant $u = \frac{\Delta x}{p_x}$ and $v = \frac{\Delta y}{p_y}$ are pixel translations for the h_{13} and h_{23} , creating a translation matrix T . Adjusting the previous homography or H_0 shifts the warped image by exactly the right number of pixels to match its true GPS displacement (T in equation 2), adjusts the translation terms while preserving the existing transformations (rotation and scale) in the homography matrix.

$$H_{adjusted} = \begin{bmatrix} 1 & 0 & u \\ 0 & 1 & v \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \quad \text{or } H_{adjusted} = T.H_0 \quad (2)$$

Algorithm 1 represents the homography initialization and adjustment process. The algorithm takes as input the UAV video frames, a reference orthomosaic, and UAV telemetry data to produce georeferenced output frames. The first frame initializes the homography through SIFT-based feature matching. For subsequent frames H_0 is recomputed when the UAV's altitude change exceeds a predefined threshold or when a significant rotational deviation is detected. Otherwise, the homography is updated using the translation matrix T . Given that DJI Mini 3 Pro SRT files record altitude changes only to one decimal place, a threshold of 0.1 m is applied to ensure reinitialization even for minor variations. This precaution prevents scaling errors, which are critical in photogrammetric workflows where accurate Ground Sampling Distances (GSDs) must be maintained.

Algorithm 1. Homography Initialization and Extension

Input: Input Frame, Reference Image, Flight Logs/Telemetry
Parameters: Altitude Threshold(m) = ± 0.1 , Rotation Threshold = $\pm 0.5^\circ$
Output: Georeferenced Output Frame
BEGIN
 For each Frame in the video:
 If first frame:
 | Initialize homography using PCA-SIFT
 | Compute control points x_a and y_a
 Else:
 If Altitude change > Threshold or Mean Rotation > Threshold (Algorithm 2)
 | Recompute homography using PCA-SIFT
 | Recompute control points x_a and y_a
 Else:
 | Rotation check using Algorithm 2
 | Readjust homography by adding translation Δx and Δy
 End

3.4. Rotation detection with ORB

The standard DJI telemetry (SRT) file contains latitude, longitude, altitude, focal length, and zoom ratio, but lacks information about the drone's compass heading. Since SIFT is computationally expensive, it is applied only to frames requiring correction due to altitude or rotational changes. Therefore, for rotation detection in consecutive frames, a faster alternative is used. Oriented FAST and Rotated BRIEF, or ORB, is a fast and efficient alternative to other algorithms in this category. It is invariant to rotation but highly sensitive to scale (Tareen and Saleem, 2018). The rotation detection occurs between consecutive frames with minimal scale variation; therefore, ORB is well-suited for this task, offering fast and reliable processing (see Algorithm 2).

Algorithm 2. Rotation Detection

Input: Current Frame, Previous Frame, ORB, and PCA-SIFT
Parameters: Rotation Threshold = $\pm 0.5^\circ$
Output: Homography Reinitialization due to Rotation
BEGIN
 If Frame_i is the initial frame:
 | Initialize homography using PCA-SIFT
 Else:
 Match ORB features between Frame_i (Current Frame) and Frame_{i-1} (Previous Frame)
 Estimate feature rotation angle (in degrees)
 If rotation angle > Rotation Threshold:
 | Reinitialize Homography with Reference Image using PCA-SIFT
 Else:
 Skip transformation for Frame_i and move to Frame_{i+1}
End

For algorithm 2, ORB is configured with a maximum of 2000 detected keypoints, while other parameters, such as a pyramid scale factor of 1.2, eight pyramid levels, a patch size of 31, and a FAST detection threshold of 20, are retained at their default OpenCV values. The rotation detection algorithm identifies and calculates keypoints and descriptors between two consecutive frames. It then applies a brute-force matcher to find the correspondences between the feature descriptors extracted. A ratio test is then performed with a threshold of 0.75, retaining only high-confidence matches. The algorithm then computes the average rotation difference between the matched keypoints to estimate the relative orientation (illustrated in Fig. 3). A threshold of $\pm 0.5^\circ$ is applied; if the rotation difference exceeds this threshold, the base homography is recalculated using the same SIFT function demonstrated in algorithm 1. The figure of $\pm 0.5^\circ$ is derived from experimentation with different drone footage at 30 FPS; a stricter threshold may be adopted to further minimize the risk of frame misalignment. The number of detected keypoints also plays a crucial role in adjusting the threshold. In the absence of explicit compass heading information, algorithm 2 assists in correcting orientation drift by detecting significant rotations and triggering the reinitialization of the H_0 SIFT, which corrects the rotation component within the homography matrix.

3.5. Accuracy assessment, sampling and moving block bootstrap

In photogrammetry, achieving accuracy is a challenging yet crucial aspect. The accuracy assessment followed the method recommended by the US National Standard for Spatial Data Accuracy (NSSDA) to measure and report the quality of geographic data (Planning, 1999). This method accounts for various uncertainties, including geodetic control coordinates, data compilation, and the final computation of ground coordinate values on the map. The assessment was conducted based on the level of conformity between the measured values and their reference values derived from the orthorectified reference image (illustrated in Fig. 4). One of the most commonly used metrics for evaluation is the RMSE (Authority, 1998). RMSE for each frame was calculated by comparing the sampled frame with the reference image. Four corners of each output frame were marked as GCPs, and the distance was calculated using the measurement tool QGIS.

$$RMSE_{frame} = \sqrt{\frac{\sum_{i=1}^n (\Delta_{GCP_i})^2}{n}} \quad (3)$$

Equation (3) represents the average deviation between the observed and a positions of GCPs within each sampled frame. Δ_{GCP_i} is positional error for the i^{th} GCP and n is the total number of GCPs in the frame. Since adjusted homography is built upon the initial homography H_0 , which means that RMSE is lowest in the frames where SIFT was applied, and could potentially increase with adjusted translation matrix. To investigate this, a relative error was calculated as $r = \frac{E - E_s}{E_s}$. Where E is the RMSE at the current frame and E_s is the RMSE where H_0 was reinitialized due to altitude change or rotational change. Translation errors were also calculated as $E - E_s$ to estimate the drift between successive $H_{adjusted}$.

Frame sampling was performed using a combined systematic and purposive scheme to ensure representative coverage. Systematic sampling selected every 50th frame and its RMSE was calculated, while purposive sampling added the nearest SIFT-corrected frame if a SIFT correction occurred between two sampled frames.

$$S = (f_k | f_k \text{ is every 50th frame or nearest to a SIFT corrected frame}) \quad (4)$$

Because $H_{adjusted}$ depends upon the nearest H_0 , frames exhibit temporal correlation. Instead of providing point estimates of georeferencing accuracy, the uncertainty within the error metrics was quantified by calculating 95 % confidence intervals (CI) for RMSE, RMSE after SIFT, translation error, and relative error. The standard confidence interval methods that assume frame-wise independence would underestimate the true uncertainty due to this temporal dependence. To address this, we employed a moving-block bootstrap (MBB) approach, which resamples contiguous blocks of frames to preserve the autocorrelation structure and produce realistic CIs (Ayinde et al., 2023). If frame-wise independence were assumed, the resulting CIs would likely appear artificially narrow, as temporal dependence inflates the actual variance (Kreiss and Paparoditis, 2011). Accounting for this dependence, therefore, yields more realistic and reliable estimates of the uncertainty. The block length L was chosen to account for temporal dependence within each

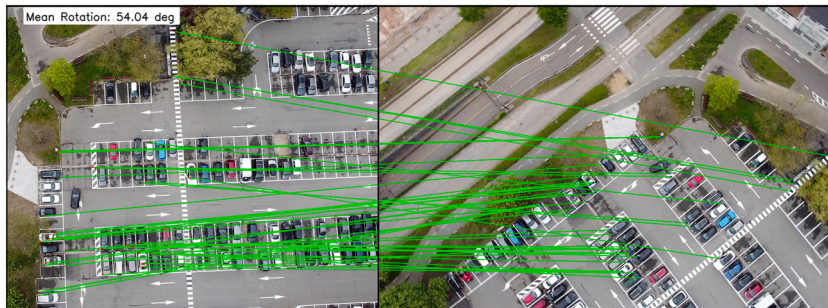


Fig. 3. Demonstration of rotation detection algorithm.



Fig. 4. An illustration of GCPs (from Reference Orthomosaic) and Mapped Points (Output Frame).

frame sequence. The initial estimate was obtained from the autocorrelation function (first lag where correlation < 0.2), and refined via block sensitivity analysis to identify the block size yielding the most stable and narrow 95 % CI for the mean. Using this block length, contiguous blocks were sampled with replacement and concatenated to form bootstrap resamples of size N . The bootstrap process was repeated 800 times, which was sufficient for the CIs to stabilize, producing reliable estimates of the mean. The 2.5th and 97.5th percentiles of the bootstrap distribution were used to define the CIs, and the analysis was performed on $n = 45$ and $n = 41$ sampled frames for Footage 01 and 02, respectively (see Table 2).

4. Results and discussion

4.1. Sampling and validation metrics

In this experiment, a total of 4200 frames from two distinct videos were georeferenced. To assess the accuracy of these georeferenced frames, at least four Ground Control Points (GCPs) were manually marked at the corners of each sampled frame. Frame sampling combined a systematic approach—selecting every 50th frame for RMSE calculation—with purposive sampling, which included the nearest SIFT-corrected frame whenever a SIFT correction occurred (see Equation (4)). This combined sampling strategy ensures coverage of both regularly spaced frames and key frames requiring homography adjustments due to altitude or rotational variations. To enhance precision, GCPs were chosen at ground level rather than elevated positions (as demonstrated in Fig. 4). These GCPs serve as known reference points, enabling the assessment of how closely the georeferencing aligns with real-world coordinates.

The accuracy of the georeferencing was quantified using three key metrics: RMSE, translation and the relative error. RMSE provides a measure of the average squared differences between the estimated values (the georeferenced points) and the actual values (GCPs). Translation error represents the cumulative drift between successive homography matrices that are not corrected by SIFT, measured in

Table 2

Accuracy metrics of experimental footage 1 and 2.

Metric	Footage	N (Frames)	Mean	SD	Median	CI 95 % Lower	CI 95 % Upper	Block Size
RMSE (cm)	Footage 01	45	95.9	44.2	94.2	81.06	114.8	2
	Footage 02	41	54.9	36.4	39.6	48.5	72.6	15
RMSE SIFT (cm)	Footage 01	45	67.5	43.3	59.8	52.8	85.7	2
	Footage 02	41	21.5	17.3	16.6	16.8	29.2	2
Translation Error (cm)	Footage 01	45	28.4	28.3	17.9	24.9	42.1	20
	Footage 02	41	33.4	30.9	24.5	26.9	43.0	15
Relative Error (unitless)	Footage 01	45	0.89	1.65	0.27	0.43	1.45	2
	Footage 02	41	1.94	1.58	1.62	1.53	2.48	15

meters, whereas the relative error, expresses how much the error varies for $H_{adjusted}$ relative to the SIFT-corrected homography. Since the homography is adjusted for each frame, the RMSE is higher in $H_{adjusted}$ compared to the control frames. Additionally, the SIFT parameters are fixed throughout the footage, therefore, the RMSE at H_o is not constant for all control frames. As the source area of keypoints changes, the resulting RMSE may also vary.

4.2. Accuracy results and error analysis

In both Footage 01 and Footage 02, the SIFT-corrected frames served as “control points”—instances when the homography is fully reset against the reference image due to a change in altitude or rotation—while for other frames, the homography is incrementally adjusted based on estimated geodesic distances Δx and Δy until the next control point. The analysis indicates that the RMSE peaks are evident in both footages, reaching 190 cm in the 900th frame of footage 1 and 175 cm in the 550th frame of footage 2 (illustrated in Fig. 5). The peak observed in footage 1 is primarily attributed to significant errors produced by the translation terms, added to the pre-existing error of H_o at the 300th frame. The H_o error is caused by changes in scene conditions such as lighting, perspective, and motion blur, which reduce the reliability of feature matches. Since the SIFT parameters are fixed prior to execution, the algorithm cannot adapt to these variations, leading to higher matching errors from frames 300–1050. While adjusting parameters such as keypoint threshold, Lowe’s ratio test, RANSAC, and FLANN can enhance performance in certain locations, it may inadvertently hinder it in others. Therefore, caution is advised when optimizing these parameters.

The experimental results show that Footage 01 exhibited higher mean RMSE of 95.9 cm which reduces to 67.5 after SIFT correction (see Table 2). In contrast, the second footage has a mean RMSE of 54.9 cm which goes down to 21.5 cm upon SIFT implementation. The mean translation error was slightly higher for Footage 02 (33.4 cm) compared to Footage 01 (28.4 cm), although both exhibited similar variability ($SD \approx 28$ –31 cm). The comparable magnitude of translation adjustments suggests that the UAV telemetry produces error at a consistent rate (other operational factors may also contribute). Furthermore, the higher translation error in Footage 02 is attributed to less frequent SIFT corrections and prolonged reliance on the drone’s telemetry. In terms of relative error, the results reveal contrasting trends between the two datasets. Footage 01 exhibited a mean relative error of 0.89, with a median of 0.27, indicating a right-skewed distribution dominated by a few large deviations. Footage 02, however, showed a higher mean relative error of 1.94 and a median of 1.62, implying that errors in Footage 02, while smaller in absolute terms, represent larger proportional deviations relative to the SIFT frames.

With respect to the CI widths, Footage 01 consistently displays broader intervals across all metrics. For example, the 95 % CI for RMSE spans 33.8 cm (81.06–114.84 cm) in Footage 01, compared to 24.1 cm (48.5–72.6 cm) in Footage 02. This broader interval is likely due to the higher degree of uncertainty introduced by SIFT, stemming from the fixed parameters and automated nature of the workflow. Whereas, the CIs for translation error are comparable, spanning 17.2 cm (24.9–42.1 cm) for Footage 01 and 16.1 cm (26.9–43.0 cm) for Footage 02.

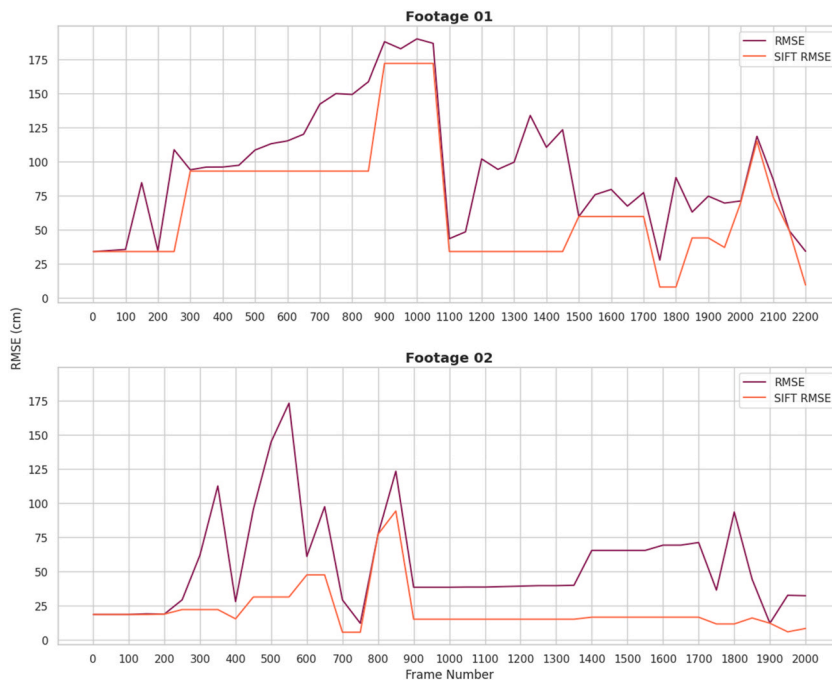


Fig. 5. Line plot with RMSE for two experimental footages error over sampled frames. The dips noticed are the anchoring frames due to SIFT implementation due to altitude change or rotation detection.

The block size estimates derived from the MBB provide insight into the degree of temporal dependence within each metric and footage. For footage 01, the smallest block size of 2 was observed for both RMSE and RMSE SIFT, as well as relative error, indicating that residuals in frame-to-frame alignment errors decorrelate rapidly. This suggests that most of the noise in the error signal of footage is short-term and driven by local fluctuations linked with randomness caused by feature matching, whereas the translation error demonstrates a larger block (block size = 20) exhibiting a temporal correlation. This longer dependence reflects the gradual propagation of displacement during incremental homography updates between control points. In Footage 02, the RMSE SIFT shows a minimal block size of 2, again indicating a low temporal dependency due to variability caused by homography corrections. However, RMSE, relative, and translation error exhibit considerably larger block sizes (15), indicating extended autocorrelation due to more stable UAV flight and fewer homography corrections.

In terms of speed, the frames with $H_{adjusted}$ took 3.847 s to process, including the rotation detection using ORB (discussed in Table 3). In contrast, after applying PCA to the descriptors, SIFT required 24 s to process a single frame. The execution time includes projection generation for each frame and exporting it as a Geotiff format, a computationally expensive task. The conventional SIFT method took 255 s to process one frame using the same footage and reference image. The time measurements are based on program execution on the Google Colab environment.

4.3. Limitations and future research implications

The primary limitation of the workflow is its dependence on reference data for feature matching, which is essential for homography initialization and corrections. If the most recent reference imagery is unavailable or the built environment has changed (e.g., due to new construction), the feature matching algorithm may not function properly. The workflow also relies on top-down or nadir images, as 2D feature matching is sensitive to angular deviations. Additionally, it does not account for terrain diversity; for instance, the orientation of the top-left building (highlighted in Fig. 6) shifts with the drone's movements because no geometric corrections or terrain relief are applied to the input frames.

Inconsistencies in telemetry, such as unnoticed altitude changes, can further increase RMSE, since homography resets are triggered by altitude variations. The automated nature of the workflow also introduces variability, as the UAV moves, the source keypoint area changes, and repeated applications of SIFT may yield different results, leading to divergent outcomes. While this experiment was conducted over relatively flat terrain, more topographically diverse environments could exacerbate parallax effects, necessitating terrain relief using a Digital Elevation Model (DEM). Future work will address these limitations while maintaining a minimal set of parameters to preserve computational efficiency.

The results indicate that the output can be integrated with downstream processes such as object detection and georegistration. The system's accuracy and scalability provide distinct advantages for specific applications, such as road traffic monitoring. In these scenarios, the proposed workflow offers spatial awareness by facilitating the analysis of aerial footage in real geographical space. Such applications can assist in vehicle position mapping, trajectory analysis, and behavioral assessment between different vehicles through aerial footage. The system's effectiveness and relevance in real-world traffic situations present significant benefits for enhancing roadway safety and traffic management.

5. Conclusions

This paper introduces a hybrid georeferencing approach that combines the feature-matching algorithm SIFT with minimal UAV metadata from a commercially available drone. The proposed method first initializes a homography matrix by matching features from the input frame with georeferenced aerial imagery. It then uses UAV telemetry to update the initialized homography based on the drone's movements with each frame. Additionally, this method compensates for the lack of information in the drone's telemetry, such as compass headings, by implementing an alternative feature-matching mechanism for detecting rotation during flight.

Experimental results demonstrate that the proposed workflow achieves sub-meter accuracy (mean RMSE ranging from 54.9 to 95.9 cm), using only a minimal set of parameters from a commercially available low-cost drone. This represents a significant advantage, as it enables automated yet precise georeferencing of drone videos without relying on expensive or customized hardware, such as RTK-enabled drones or high-end GNSS/IMU systems. The hybrid method also provides substantial computational gains; for the experimental dataset, it was approximately 67 times faster than conventional SIFT and over 6 times faster than PCA-SIFT, highlighting its efficiency for practical applications. Moreover, the workflow is modular and flexible—SIFT can be replaced with alternative feature-matching approaches, including deep learning-based methods, without modifying the core telemetry integration logic. This design inherently future-proofs the system, allowing immediate integration of faster or more robust algorithms as they become available,

Table 3
Average processing time per frame for different feature-matching methods.

#	Method ^a	Average Time per Frame (s) ^b
1	Conventional SIFT	255
2	PCA-SIFT	24
3	Homography adjustment and ORB rotation check	3.8

^a Script was run on the default CPU for Colab i.e. an Intel Xeon CPU with 2 vCPUs (virtual CPUs) and 13 GB of RAM.

^b Average time includes projection definition and geotiff export.

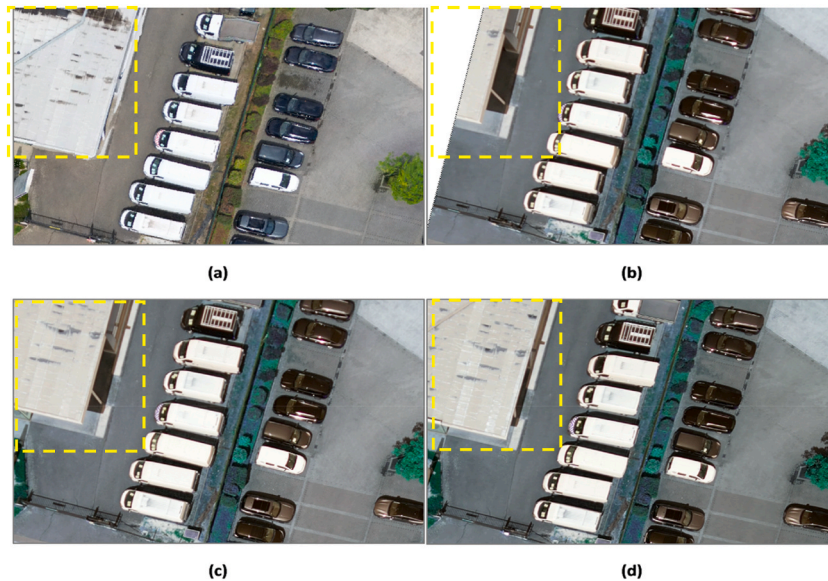


Fig. 6. (a) shows the orthorectified reference image, whereas (b), (c), and (d) are output frames from the proposed method. There is a noticeable perspective shift with UAV movement.

while still leveraging minimal UAV metadata. Future research will explore deep learning-based alternatives, evaluate different drone hardware, and test the workflow under more diverse terrain conditions to enhance both computational efficiency and georeferencing accuracy.

CRediT authorship contribution statement

Muhammad Waqas Ahmed: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Muhammad Adnan:** Supervision, Resources, Project administration. **Muhammad Ahmed:** Supervision, Project administration. **Davy Janssens:** Resources, Project administration, Funding acquisition. **Geert Wets:** Resources, Project administration, Funding acquisition. **Afzal Ahmed:** Project administration. **Wim Ectors:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Data availability statement

The data presented in this study are available on request from the corresponding author due to privacy concerns.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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